University of Zagreb
Faculty of Science
Department of Mathematics

Duje Perković, Matko Petričić, Paula Horvat

Anomaly Detection in Images Using Diffusion Maps

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Contents

Abstract

In this paper, we develop a method for detecting anomalies in images based on diffusion maps for dimensionality reduction and hierarchical processing using a Gaussian pyramid. We first construct a kernel matrix and compute a diffusion map on a subsample of pixels, using a global density parameter σ determined from the empirical variance of distances between randomly selected pairs of pixels. The resulting embedding is extended to the remaining pixels using the Nyström extension, which preserves the geometric structure of the data set while significantly reducing computational complexity.

For each pixel at every level of the pyramid, we define an anomaly index that measures how isolated the pixel is in the new coordinates. By introducing an inner mask, we avoid underestimating spatial anomalies that would otherwise be hidden among their immediate neighbors.

To apply the method efficiently at high resolutions, we build a Gaussian pyramid and begin detection at the lowest-resolution level G_L. Suspicious pixels are propagated and supplemented with random samples when moving to each finer level, until the original level G_0 is reached. This process substantially accelerates execution without sacrificing detection accuracy. Experimental examples show reliable isolation of anomalies with a considerable reduction in runtime, owing to the sparse k-nearest-neighbor implementation and the Nyström extension. Further refinements may include automatic parameter selection and threshold optimization for different types of anomalies.

1 Introduction

In this paper, we develop a method for detecting anomalies in images based on diffusion maps. We consider anomalies to be significant deviations from the periodic structure of the background or from the dominant pattern in the image. Anomaly detection involves the following challenges:

- a large amount of data in a single image,
- the influence of noise, which can lead to incorrect classifications,
- a lack of anomaly examples for testing despite an abundance of normal data,
- the dispersion of normal patterns, such as the background, across multiple clusters.

Many approaches in the literature are based on statistical methods, but their success depends strongly on the precise choice of distribution parameters, which is itself a complex problem. Our approach does not assume a known data distribution and does not require a training phase. It is necessary to define a set of features and a distance measure for comparing the local similarity of samples. As optional input, one may also specify the expected spatial size of anomaly regions in the image. We expect anomalies to lie in a separate, lower-density region in the transformed data, while normal data are more densely concentrated in another region.

In this work, we use diffusion maps as a dimensionality reduction method. The most computationally demanding step is constructing the kernel matrix, since it requires computing the distance between every pair of samples. The computational burden can be reduced by selecting a subset of samples and extending the results to the rest of the data set using *out-of-sample* methods, which are often used when processing high-dimensional images.

An additional speedup is obtained by constructing a *sparse* kernel matrix using a k-nearest-neighbor search: instead of all pairs, the kernel is computed only between each sample and its nearest neighbors. This produces a *sparse* matrix, and its spectral decomposition becomes considerably more efficient thanks to algorithms designed for *sparse* data.

2 Definitions and Main Ideas

2.1 Diffusion Maps

Diffusion maps address the problem of nonlinear dimensionality reduction: an initial data set $S \subset \mathbb{R}^N$ is mapped into a lower-dimensional space $\mathbb{R}^d$.
For a parameter $\sigma > 0$ and $S = \{x_1, \ldots, x_n\}$, we define the kernel $K \in M_n(\mathbb{R})$ by

$$K(i,j) = e^{\frac{-||x_i - x_j||^2}{\sigma}}.$$

The idea is to consider a graph whose vertices are labeled by the elements of S, where the weight of the edge between vertices x_i and x_j is given by $K(i,j)$. Notice that the matrix K is positive and symmetric.
With the normalization

$$q(i) = \sum_{j=1}^{n} K(i,j),$$

$$Q(i,j) = K(i,j)/q(i),$$

we obtain a row-stochastic matrix $Q \in M_n(\mathbb{R})$. Notice that 1 is an eigenvalue with associated eigenvector $(1,\ldots,1) \in \mathbb{R}^n$, and since $||Q||_\infty = 1$, we have $r(Q) = 1$, where r denotes the spectral radius.
The matrix Q has a Markov-chain interpretation: $Q(i,j)$ represents the transition probability from element x_i to element x_j. Raising Q to a power $t \in \mathbb{N}$ represents the action of the Markov chain after t steps.
The matrix Q is obtained as $Q = D^{-1}K$, where $D = \operatorname{diag}(q(1),\ldots,q(n))$. Observe that

$$Q = D^{-1/2}(D^{-1/2}KD^{-1/2})D^{1/2}.$$

Thus, Q is similar to the symmetric matrix $D^{-1/2}KD^{-1/2}$, so it has real eigenvalues and is diagonalizable. Let $|\lambda_1| \geq \cdots \geq |\lambda_n|$ denote the eigenvalues, and let $\phi_1,\ldots,\phi_n$ be the corresponding eigenvectors of $D^{-1/2}KD^{-1/2}$. We define $\Theta_t : S \to \mathbb{R}^d$ by

$$\Theta_t(x_i) = (q(i)^{-1/2}\lambda_1^t\phi_1(i),\ldots,q(i)^{-1/2}\lambda_d^t\phi_d(i)).$$

The main motivation for defining this map comes from the definition of diffusion distance:

$$\mathscr{D}_t(x_i,x_j)^2 = \sum_{l=1}^{n} (Q^t(x_i,x_l) - Q^t(x_j,x_l))^2/q(x_l).$$

Indeed, a result from the lectures gives

$$\mathscr{D}_t(x_i,x_j)^2 = ||\overline{\Theta}_t(x_i) - \overline{\Theta}_t(x_j)||^2,$$

where $\overline{\Theta}$ is defined using the full spectral decomposition of $D^{-1/2}KD^{-1/2}$. The resulting set has Euclidean distances equal to diffusion distances. Since diffusion distances take all points of the original set into account, the transformed set captures the **global structure** of the original data.

In general, because the eigenvalues decay rapidly, one can take $d << n$ and still obtain an accurate approximation of the full diffusion map $\overline{\Theta}$.

In the context of anomaly detection, we expect anomalies among all pixels in an image to have a clear geometric structure that will be well separated after applying the diffusion map. The application itself is explained later.

2.2 Choosing the Parameter σ

The initial graph construction highlights the importance of choosing σ well when constructing the kernel K. If σ is too small, the graph has weakly connected vertices; if it is too large, all vertices become strongly connected to one another.

In other words, when σ is too small, it is difficult to determine what is anomalous because there are too many candidates; when σ is too large, the anomaly is too strongly connected to the other normal pixels and is difficult to recognize. Instead of using exactly one value for σ, we can assign different values depending on the points x_i, x_j for which the weight $K(x_i,x_j)$ is computed.

One method is to assign each point x_i a value σ_i, defined using one of its neighbors $u_1,\ldots,u_{n-1}$, where $n = \text{card}(S)$ and the neighbors are ordered from nearest to farthest. For the l-th nearest neighbor, where $l \in \{1,\ldots,n-1\}$, we can set

$$\sigma_i^2 = ||x_i - u_l||^2$$

and define

$$K(x_i, x_j) = e^{\frac{-||x_i - x_j||^2}{\sigma_i \sigma_j}}.$$

This is intuitively meaningful because we expect the densities around isolated anomaly pixels and normal pixels to differ substantially.

2.3 Extending Diffusion Maps

2.3.1 Partial Diffusion Map

For large data sets, computing the kernel matrix K and the spectral decomposition of $D^{-1/2}KD^{-1/2}$ is expensive. It is therefore desirable to construct a diffusion map on a smaller number of data points and then extend it. A common method is the Nyström extension.

Let $S = \{x_1, \ldots, x_n\}$ be a data set and let $M = \{x_1, \ldots, x_s\}$ be a subset with $s < n$. We define $\widetilde{K}$ as the upper $s \times n$ submatrix of $K \in M_n(\mathbb{R})$. We define $\widetilde{D}$ as an $s \times s$ diagonal matrix such that $\widetilde{D}(i,i) = \sum_{k=1}^{n} \widetilde{K}(i,k)$ for all $i \in \{1, \ldots, s\}$. Finally, we define

$$\widetilde{A} = \widetilde{D}^{-1/2}\widetilde{K}D^{-1/2},$$

where D is an $n \times n$ diagonal matrix containing the column sums of the newly defined $\widetilde{K}$. Instead of a spectral decomposition, we compute an SVD. Equivalently, for the symmetric matrix $\widetilde{A}\widetilde{A}^\top$, we find the representation $\widetilde{\phi}\Sigma^2\widetilde{\phi}^\top$, where $\Sigma = \operatorname{diag}(\sigma_1, \ldots, \sigma_s)$. We then define the **partial diffusion map** $\widetilde{\Theta} : M \to \mathbb{R}^d$ by

$$\widetilde{\Theta}(x_i) = (\widetilde{D}(i,i)^{-1/2}\sigma_1\widetilde{\phi}_1(i), \ldots, \widetilde{D}(i,i)^{-1/2}\sigma_s\widetilde{\phi}_s(i)).$$

For this map, if Θ denotes the full diffusion map on all of S, we obtain

$$||\widetilde{\Theta}(x_i) - \widetilde{\Theta}(x_j)|| = ||\Theta(x_i) - \Theta(x_j)||, \quad \forall x_i, x_j \in M.$$

Thus, the structure of the mapped subset M is preserved as if the full diffusion map had been computed.

In practice, as with the usual diffusion map, we do not use all eigenvectors, but only the first few whose associated eigenvalues are relevant, meaning sufficiently large.

2.3.2 Nyström Extension

Given the subset M and its complement $\overline{M}$, the matrix $A = D^{-1/2}KD^{-1/2}$ can be written in block form as

$$A = \begin{bmatrix} A_{(M,M)} & A_{(M,\bar{M})} \\ A^T_{(M,\bar{M})} & A_{(\bar{M},\bar{M})} \end{bmatrix},$$

where $A_{(X,Y)}$ denotes the part of the matrix containing the scaled edge weights between vertices from the sets X and Y. With this notation,

$$\widetilde{A} = \begin{bmatrix} A_{(M,M)} & A_{(M,\bar{M})} \end{bmatrix}.$$

The question is therefore how well we can approximate $A_{(\bar{M},\bar{M})}$.

We expect the subset M to be selected so that normal pixels and anomaly pixels appear in the same proportions as in the original data set. Then the diffusion-distance information contained in $\widetilde{A}$ should be sufficient for approximating the remaining diffusion distances. In terms of numerical matrix rank, meaning sufficiently large singular values, we would expect $\mathrm{numR}(A) \approx \mathrm{numR}(\widetilde{A})$.
Using the spectral decomposition, write

$$A = \begin{bmatrix} U_1 \Lambda U_1^\top & U_1 \Lambda U_2^\top \\ U_2 \Lambda U_1^\top & U_2 \Lambda U_2^\top \end{bmatrix}.$$

Since $A_{(\bar{M},M)} = U_2 \Lambda U_1^\top$, we obtain $U_2 = A_{(\bar{M},M)} U_1 \Lambda^{-1}$. Substituting this into $A_{(\bar{M},\bar{M})}$, we get

$$A_{(\bar{M},\bar{M})} = A_{(\bar{M},M)} U_1 \Lambda^{-1} \Lambda \Lambda^{-1} U_1^\top A_{(M,\bar{M})},$$

that is,

$$A_{(\bar{M},\bar{M})} = A_{(\bar{M},M)} U_1 \Lambda^{-1} U_1^\top A_{(M,\bar{M})}.$$

Therefore, we approximate this expression by

$$A_{(\bar{M},\bar{M})} \approx A_{(\bar{M},M)} A^{-1}_{(M,M)} A_{(M,\bar{M})}.$$

In this way, we construct an approximation $\hat{A}$ of A and use it to define the so-called **orthogonal Nyström extension**. More precisely, we seek a decomposition $\hat{A} = \hat{\phi} \Lambda \hat{\phi}^\top$, where $\hat{\phi} \in M_{n \times s}(\mathbb{R})$ has orthonormal columns and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_s)$ is an $s \times s$ diagonal matrix.

We then define $\hat{\Theta}(x_i) : S \to \mathbb{R}^s$ by

$$\hat{\Theta}(x_i) = (D(i,i)^{-1/2}\lambda_1\hat{\phi}_1(i), \ldots, D(i,i)^{-1/2}\lambda_s\hat{\phi}_s(i)).$$

For the map obtained in this way, the structure of the mapped subset M is preserved as if the standard diffusion map had been applied:

$$\forall x_i, x_j \in M, \quad ||\hat{\Theta}(x_i) - \hat{\Theta}(x_j)|| = ||\bar{\Theta}(x_i) - \bar{\Theta}(x_j)||.$$

2.4 Gaussian Pyramid

In anomaly detection, and in any image-based object recognition task, every pixel matters. Each pixel contributes to the larger image that carries information. However, an image of resolution 200×200 contains 40000 pixels, so working with such an amount of data can be computationally demanding.

Therefore, as mentioned in the previous section, it is practical to replace a large data set with a smaller subset. More precisely, we compute the diffusion map on a subset of the initial data and then approximate the full data set using the result. It is clear that the overall result then depends on the initially selected subset. Thus, if the anomaly was not at least partially present in the initial data set, the diffusion-map values will not capture the nature of the anomaly. In other words, all anomalies or data points that are far from the test set will not be extended to appropriate coordinates representing their distance from the test set. For this reason, we cannot rely on the described method alone: the results are strongly conditioned by the initial random selection of data. In Figure 2.1, part a) shows the original high-resolution image in which the anomaly is being searched for. The upper row shows the detection result obtained with a random choice of the initial data subset that clearly does not contain anomalous data, so the final result is incorrect. The second row shows detection when the initial subset did contain anomalous data, and the algorithm successfully completed the task.

Another possible approach is to perform detection on a lower-resolution version of the image. The advantage of using a coarse resolution is that a larger percentage of the samples can be used because the image is smaller, making it more likely that the anomaly will be selected in the initial subset. The drawback is that the chosen resolution may limit the ability to detect small anomalies. Also, because fine details are blurred, the anomaly may be less

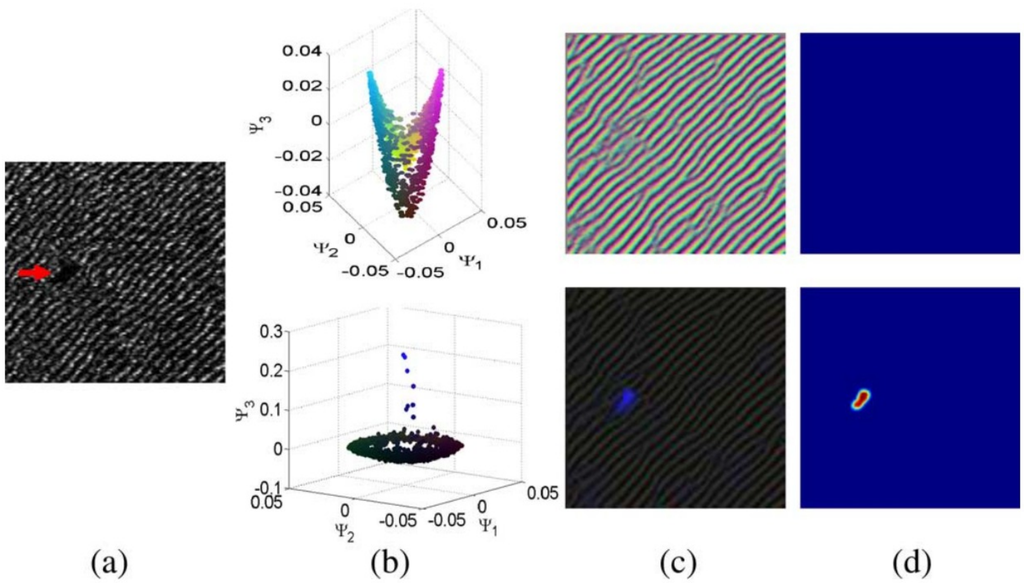

Figure 2.1: Dependence of anomaly detection on the initial data subset

pronounced relative to the background. This requires lowering the detection threshold, which results in a larger number of false alarms. In Figure 2.2, the original images are shown in the first row, while the anomaly-index value is shown in the lower row. Part a) shows anomaly detection on the image with the best resolution, where the best result is also obtained. Already in part b), detection is poor and several false positives appear in the image. In case d), detection does not occur at all.

Given the advantages and disadvantages of both methods, a natural question arises: can we design an approach that combines their strengths while reducing their weaknesses? The method implemented in this paper is motivated precisely by this idea.
Our method aims to reduce computational complexity, that is, runtime, while simultaneously improving detection results. To overcome the limitations of randomly choosing the initial subset of the image on which the diffusion map is computed, we perform detection at multiple resolutions of the original image. We assume that anomalies in the image are larger than one pixel, so they can be detected at several resolution levels. At lower resolution, it is computationally feasible to sample a larger percentage of the image, so anomaly detection at that level is less susceptible to errors caused by a missing representation of anomaly pixels in the initial subset. Therefore, we want to use the advantages of anomaly detection at different scales to overcome the disadvantages of random initial subset selection, that is, sampling.

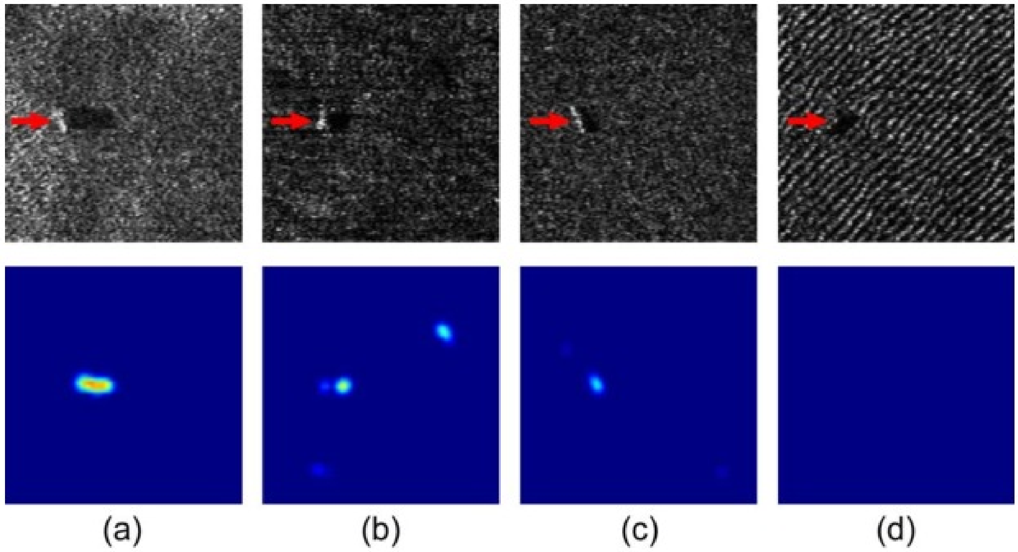

Figure 2.2: Dependence of anomaly detection on image resolution

Because our method performs anomaly detection at several image-resolution levels, even if an anomaly is missed at a coarser level, for example because it is not sufficiently emphasized there, it can still be detected at one of the subsequent finer levels. In addition, at coarse levels it is possible to lower the anomaly-detection threshold, since this will not increase the final number of false alarms: the final decision is made only at full resolution. In this way, anomalies can be detected at higher levels even at the cost of more false alarms, because those false alarms are removed in the final stage.

We now have the motivation to present our method. The approach is based on constructing a Gaussian pyramid representation of the image. Starting from the coarsest, least detailed resolution, a diffusion map is constructed based on a subset of the data. Since the image at that level is smaller, it is possible to sample a larger percentage of the image for building the diffusion map, sometimes even all pixels. Then an anomaly score is computed for each pixel in order to determine which pixels are suspected to be anomalous at that level. These pixels are stored and used as part of the sample that enters the algorithm for creating the diffusion map at the finer resolution. The remaining pixels used by the algorithm are selected randomly from the image. This process continues from level to level, where each previous level provides prior information about which pixels from the data set should be used to construct the diffusion map in a meaningful way. Suspicious pixels from the previous level are always included in the sample used to construct the diffusion map at the next level. This approach significantly increases the effectiveness of a diffusion-based anomaly detector.

3 Implementation

3.1 Algorithm

How should a representative sample of pixels be selected? A larger number of pixels increases precision but slows the algorithm down, while a smaller number accelerates execution at the cost of lower precision. The idea is to exploit the advantages of both approaches: when we already use a large number of pixels, we choose them in a meaningful way.

For a given image I, we build a Gaussian pyramid $(G_l)_{l=0}^{L}$ with L levels. Starting from the top $G_0 = I$, each subsequent level G_{l+1} is obtained by reducing the resolution of the previous level G_l. A *Gaussian low-pass filter* is applied to the image G_l, and then the width and height are reduced by a factor of 2. If the level is too high, the image G_l becomes blurred, and a larger number of randomly selected pixels can be used, sometimes even all pixels. Because of the blurring caused by the filter, anomalies partially merge with the background, so the algorithm will mark many pixels at that level as anomalous. However, if the level is not reduced too aggressively, true anomalies will still be detected. We call all pixels marked by a level as potential anomalies *suspicious pixels*.

After constructing the Gaussian pyramid, we begin detection on the lowest-resolution image G_L, where suspicious pixels are marked. When moving to the higher level G_{L-1}, we expand this set by randomly selecting additional pixels, while maintaining an acceptable runtime. This iterative procedure is repeated throughout the pyramid, propagating suspicious pixels from level to level until we obtain the final anomaly map at level G_0.

At each level, we construct a kernel matrix for the given subset of pixels. To reduce computation time, we use k-nearest neighbors: a vertex x_i is connected to a vertex x_j if x_i is among the k nearest neighbors of x_j; otherwise, we set $w(x_i, x_j) = 0$. This gives a sparse matrix and further accelerates the algorithm. It remains to define: *How do we determine whether a pixel is anomalous at level G_l?*

3.2 Anomaly Index

At each level G_l, after computing the diffusion map, the *anomaly index* $C_l(i)$ is computed for every pixel i. Pixels whose anomaly index lies above the 95th percentile are considered *suspicious*. If the image contains an anomaly, then, compared with the rest of the image, it will have a high anomaly index and will be sampled more densely at the next level. It remains to explain how the anomaly index is computed.

For a pixel i, the distance to a pixel j is defined by

$$\overline{w}(i,j) = e^{-\frac{\|\Psi(x_i)-\Psi(x_j)\|_2^2}{\overline{\sigma}}} \in (0,1].$$

Here the parameter $\overline{\sigma}$ is a global property of the entire sample, unlike the previous local choice σ_i, where each pixel had approximately K neighbors. Now we want the opposite: to introduce a single density parameter, because we assume that anomalies occur in isolated regions, significantly far from the cluster of normal pixels.

Following this intuition, we choose $\overline{\sigma}$ as follows. We randomly select n_{pair} pairs of pixels and compute the distance $\|\Psi(x_i) - \Psi(x_j)\|_2$ for each pair. Since most such pairs consist of normal pixels, their empirical variance, denoted by σ^2_{pair}, reflects the dispersion of the data set without anomalies well. Finally, we set

$$\overline{\sigma} = r\sigma^2_{\text{pair}},$$

where the parameter r controls the sharpness of the exponential function and thereby determines how close we want normal pixels to be.
We introduce several additional parameters and notations:

- $W \in \mathbb{N}$: the half-width of the square neighborhood, so N_i is a square of side length $2W+1$ centered at i.
- $m = |N_i|$.
- $M_{\text{mask}} \in \mathbb{N}$: the side length of the inner concentric square mask, denoted by M_i.
- $S_i = \{\, j \in N_i : M_{\text{mask}} \le d(i,j) \le W\}$, that is, the set of neighbors outside the mask but inside the window.

We can now define the anomaly index for pixel i at level l:

$$C_l(i) = 1 - \frac{1}{m}\sum_{j\in S_i} \overline{w}(i,j).$$

The reason for masking the inner pixels of the neighborhood is that we do not want to compare a pixel with its immediate neighbors. We assume that the anomaly is spatially relatively large. If we used a window that was too small, a pixel inside an anomalous region would be hidden among neighboring pixels that are similar in diffusion coordinates, and it would receive an anomaly score that is too low. We therefore exclude the inner mask to force the model to consider only more distant pixels, which allows it to detect isolated anomalies correctly.

The sum

$$\sum_{j \in N_i} \overline{w}(i, j)$$

can be interpreted as the number of *close* neighbors of point i within the window N_i, where closeness is determined by diffusion distance. Pixels with few close neighbors in the new coordinates, that is, pixels isolated relative to the rest of the image, receive a very high anomaly index, while pixels with many similar neighbors, and hence low diffusion distance, remain at a low index. The window size and inner mask should be chosen according to the expected spatial size of the anomalies and the specifics of the application.

4 Results

Before running the algorithm, it is necessary to estimate at least approximately the size of the anomaly in the image being processed. This is done manually until optimal results are obtained.

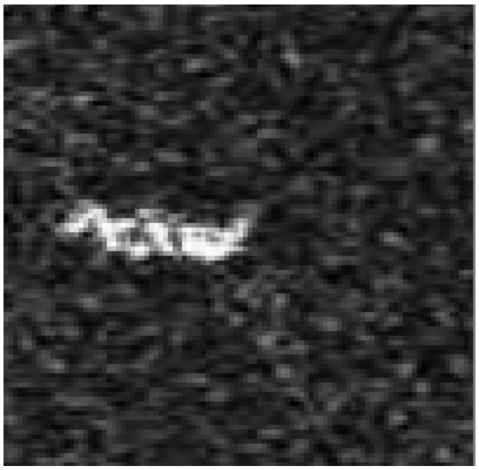

Figure 4.1: Original image

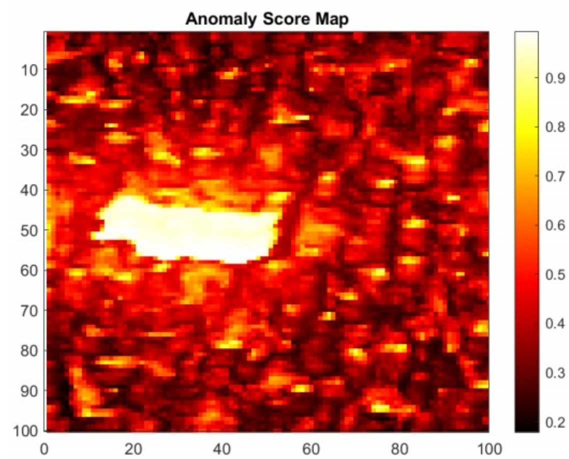

Figure 4.2: Pixel-wise anomaly score

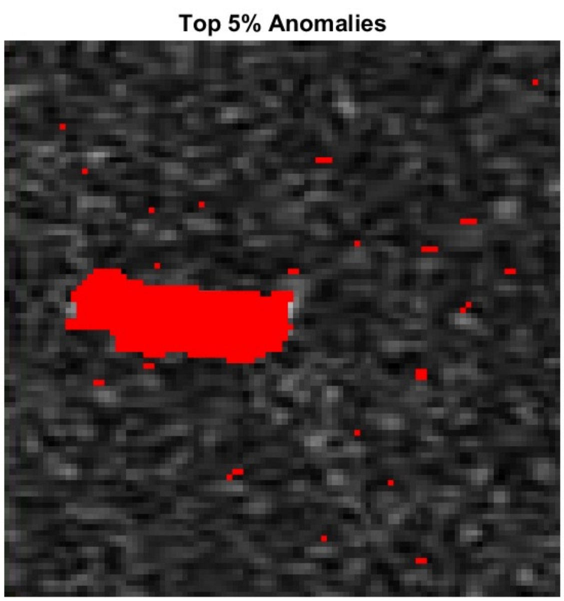

Figure 4.3: 5% of pixels with the highest anomaly score

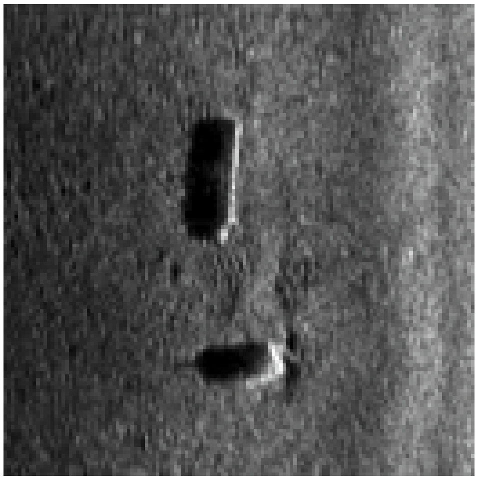

Figure 4.4: Original image

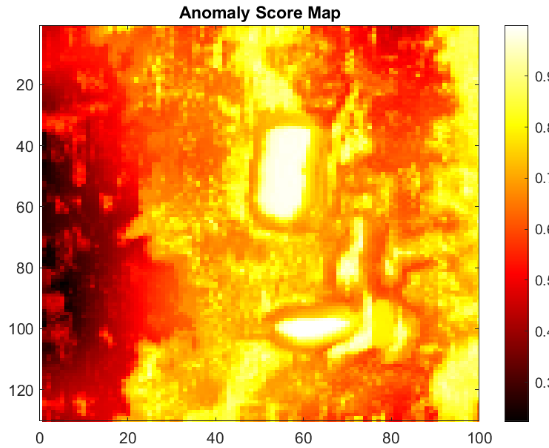

Figure 4.5: Pixel-wise anomaly score

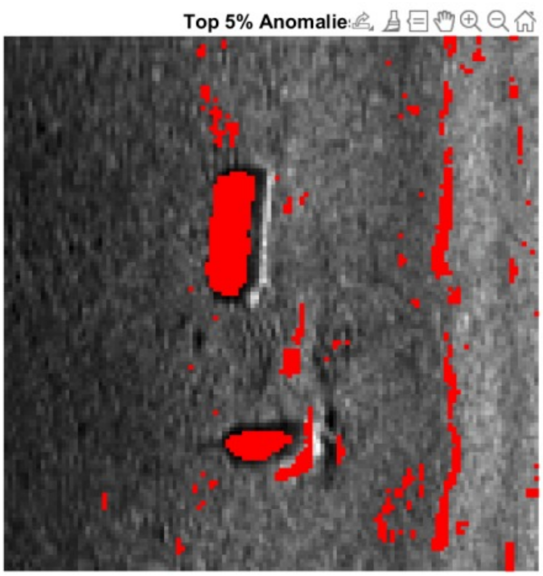

Figure 4.6: 5% of pixels with the highest anomaly index

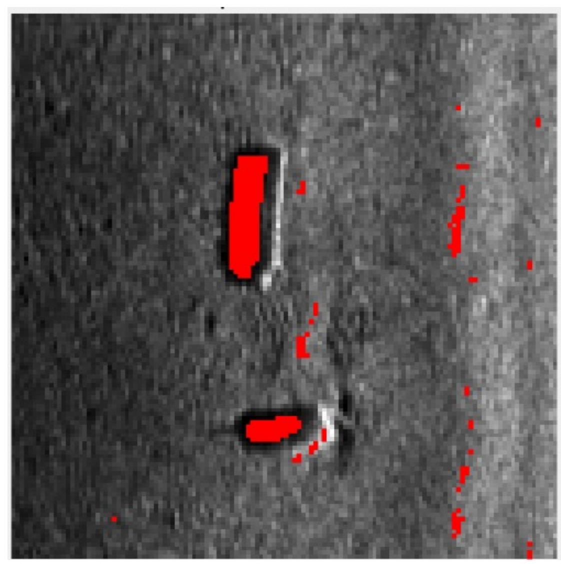

Figure 4.7: 2% of pixels with the highest anomaly index

5 Conclusion

In this paper, we presented a method for automatic anomaly detection in images that combines diffusion maps with hierarchical processing through a Gaussian pyramid. We first constructed a diffusion map on a subsample of pixels using a local density parameter based on the K nearest neighbors. The resulting coordinates were extended to the entire image using the Nyström approach, preserving the geometric structure of the data with significantly lower computational complexity.

For each level of the pyramid, we computed the anomaly index $C_l(i)$, which measures the sum of exponentially damped diffusion distances from neighbors outside the inner mask. By masking immediate neighbors, we prevented the underestimation of spatially connected anomalies. We then performed detection iteratively: at the coarsest level G_L, we marked suspicious pixels, and when moving to a finer level, we supplemented them with random samples until the final anomaly map was obtained at level G_0.

The experiments showed that our approach successfully isolates anomalies while substantially reducing runtime, owing to the K-nearest-neighbor method and the Nyström extension. Future work could automate the selection of parameters, such as $\overline{\sigma}$, neighborhood size, and inner-mask size, and investigate adaptive thresholds for different types of anomalies, as well as applications in real-time systems and video sequences.

6 References

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